



Python Coding Questions



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Basic Level

EVEN OR ODD

```
# Check if a number is even or odd
num = int(input("Enter a number: "))
print("Even" if num%2 == 0 else "Odd")
```

REVERSE A STRING

```
# Reverse a string using slicing
s = "Python"
print(s[::-1]) # Output: "nohtyP"
```

USING RANGE

```
# Print numbers from 1 to 10
for i in range(1, 11):
    print(i, end=" ")
```

FIND FACTORIAL

```
# Using built-in function for factorial

import math
num = 5
print(math.factorial(num)) # Output: 120
```

CHECK LEAP YEAR

```
# Check if a given year is a leap year
year = int(input("Enter a year: "))
if (year % 4==0 and year % 100!=0) or (year % 400==0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```



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Medium Level

●●● LIST COMPREHENSION

```
# list of squares using list comprehension
squares = [x**2 for x in range(1, 11)]
print(squares) # Output: [1, 4, 9, 16, 100]
```

●●● STRING DIGITS - INTERGER

```
# Convert string "12345" into a list of integers
s = "12345"
num_list = list(map(int, s))
print(num_list) # Output: [1, 2, 3, 4, 5]
```

●●● CHECK IF PALINDROME

```
# Function to check if a string is a palindrome
def is_palindrome(s): #Same forward and backward
    return s == s[::-1]
print(is_palindrome("madam")) # Output: True
```

●●● GENERATE FIBONACCI SERIES

```
# Each number is the sum of the previous two
n = int(input("Enter the number of terms: ")) #10
a, b = 0, 1

for _ in range(n): # _ when loop variable is not needed
    print(a, end=" ")
    a, b = b, a + b
# Output - 0 1 1 2 3 5 8 13 21 34
```

●●● CHECK IF ARMSTRONG NUMBER

```
def is_armstrong(n): # Sum of powered digits
    num_str = str(n) # convert number to string first
    power = len(num_str)
    return n==sum(int(digit) ** power for digit in num_str)
print(is_armstrong(153)) # Output: True
```



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Advanced Level



USE LAMBDA FUNCTION

```
# Lambda function to compute square of a number
square = lambda x: x ** 2
print(square(5)) # Output: 25
```



USE MAP() FUNCTION

```
# Using map() to double the values in a list
nums = [1, 2, 3, 4]
print(list(map(lambda x: x * 2, nums)))
# Output: [2, 4, 6, 8]
```



USE FILTER() FUNCTION

```
# Filter even numbers from a list using filter()
nums = [1, 2, 3, 4, 5, 6]
print(list(filter(lambda x: x % 2 == 0, nums)))
# Output: [2, 4, 6]
```



REMOVE DUPLICATES FROM LIST

```
# Remove duplicates from a list

nums = [1, 2, 2, 3, 4, 4, 5]
print(list(set(nums)))

# Output: [1, 2, 3, 4, 5]
```



COUNT VOWELS IN A GIVEN STRING

```
# Function to count vowels in a given string
def count_vowels(s):
    return sum(1 for char in s.lower() if char in "aeiou")

print(count_vowels("Hello World"))
# Output: 3
```



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Expert Level

● ● ● PYRAMID PATTERN

```
n = 5
for i in range(1, n + 1):
    print(" " * (n - i) + "*" * (2 * i - 1))
```

● ● ● FIND DUPLICATES IN A LIST

```
duplist = [1,2,3,4,4,5,5]
unique_list = list(set(duplist))
print(unique_list)
# Output - [1,2,3,4,5] but in a random order
```

● ● ● MISSING NUMBER IN A LIST

```
def find_missing(nums, n):
    return n * (n + 1) // 2 - sum(nums)
print(find_missing([1, 2, 4, 5], 5))
# Output: 3
```

● ● ● SECOND LARGEST ELEMENT

```
# Find the second largest number in a list
def second_largest(nums):
    nums = list(set(nums)) # Remove duplicates
    nums.sort(reverse=True)
    return nums[1] if len(nums) > 1 else None
print(second_largest([10, 20, 40,])) # Output: 20
```

● ● ● TWO SUM PROBLEM

```
# Find two numbers that add up to a target sum
def two_sum(nums, target):
    seen = {}
    for i, num in enumerate(nums):
        complement = target - num
        if complement in seen:
            return [seen[complement], i] # Return indices of the two numbers
        seen[num] = i
    return []
print(two_sum([2, 7, 11, 15], 9)) # Output: [0, 1]
```



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